

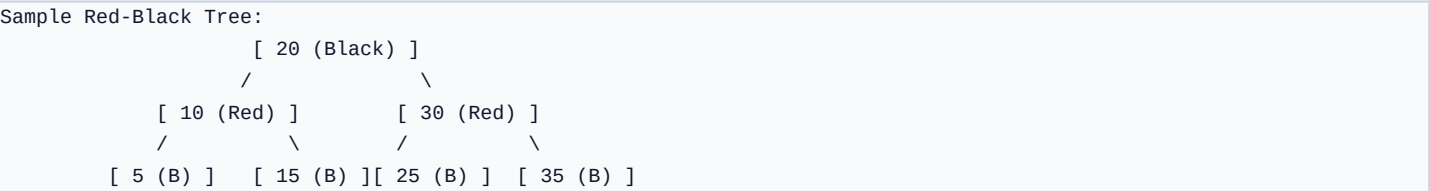
Self-Learning: Red-Black Trees -- Data Structures

Definition: A Red-Black Tree is a self-balancing Binary Search Tree where each node stores an extra bit representing color (Red or Black), satisfying specific balance constraints that ensure the tree height remains bounded by $2 \log_2(N + 1)$.

1. Detailed Technical Explanation

The 5 Red-Black Tree Properties:

1. Node Color: Every node is either RED or BLACK.
2. Root Color: The root node must be BLACK.



2. AVL Tree vs Red-Black Tree Comparison

Feature	AVL Tree	Red-Black Tree		
Balance Strictness	**Strictly Balanced** (**	BF	≤ 1**). Height **pprox 1.44 \log	**Loosely Balanced**. Longest path at
Search Speed	Faster lookups due to strictly sn	Slightly slower search than AVL		
Insertion / Deletion Speed	Slower (frequent rebalancing ro	**Faster insertions/deletions** (		
Use Cases	Search-intensive workloads (Lo	Standard library maps (C++ `std		

3. Insertion in Red-Black Tree

When inserting a new key:

1. Insert the node as a new leaf, coloring it RED.

4. Quick Recall Flow

RB-Tree -> 5 Rules: Root Black, Leaves Black, No Consecutive Reds, Uniform Black-Height -> Color Flips & Rotations